Decentralized Finance

Payment Systems

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Do you want to innovate?

- What is the problem that is being solved?
- What is the expense/inefficiency in the legacy system?

Consumer Payments in the US

18th Century

1950

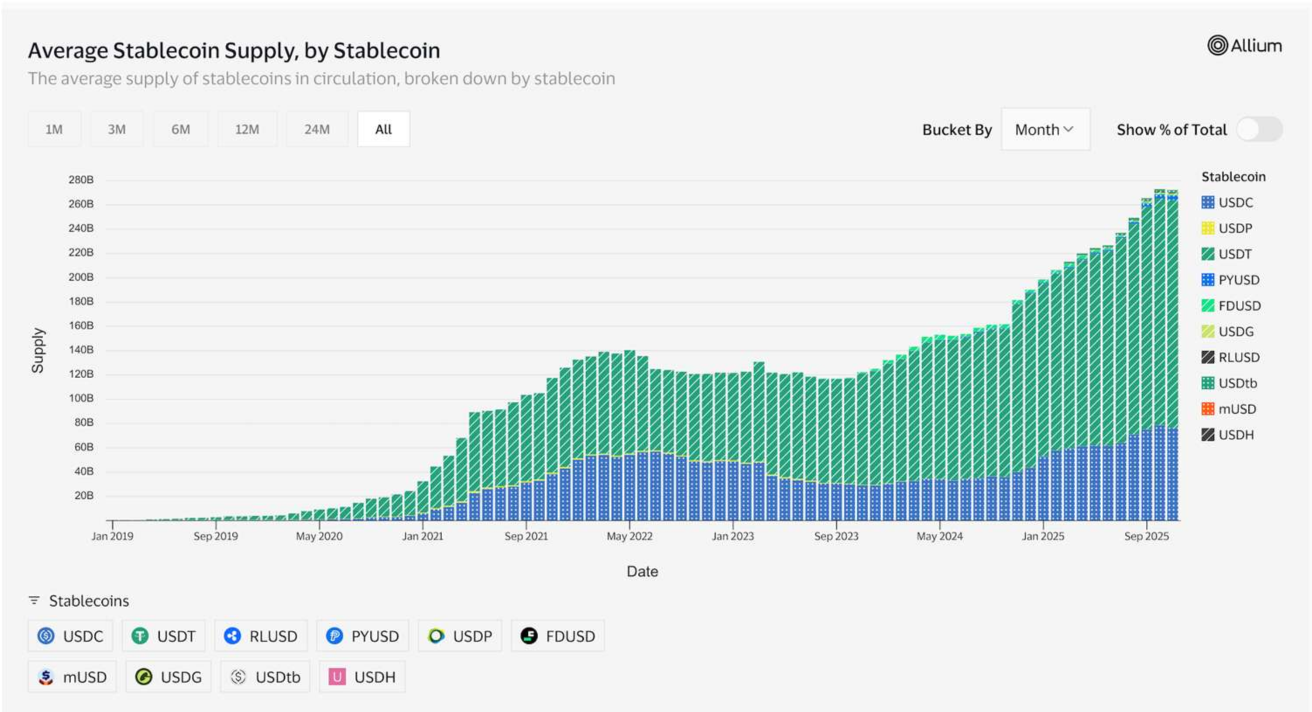
1998

2009

2017

2025

Circulating Supply



Source: Visa

USDT

Bitfinex and Tether Fined $18.5M in Settlement With NY Attorney General, Both Firms Barred From Trading in the City

Tether Launching USAT

From a consumer's point of view

- Many payment methods (free)

1. Credit/Debit cards
2. Zelle
3. Paypal
4. Money Market Funds

- Costs incurred by businesses

Money is a Social Institution

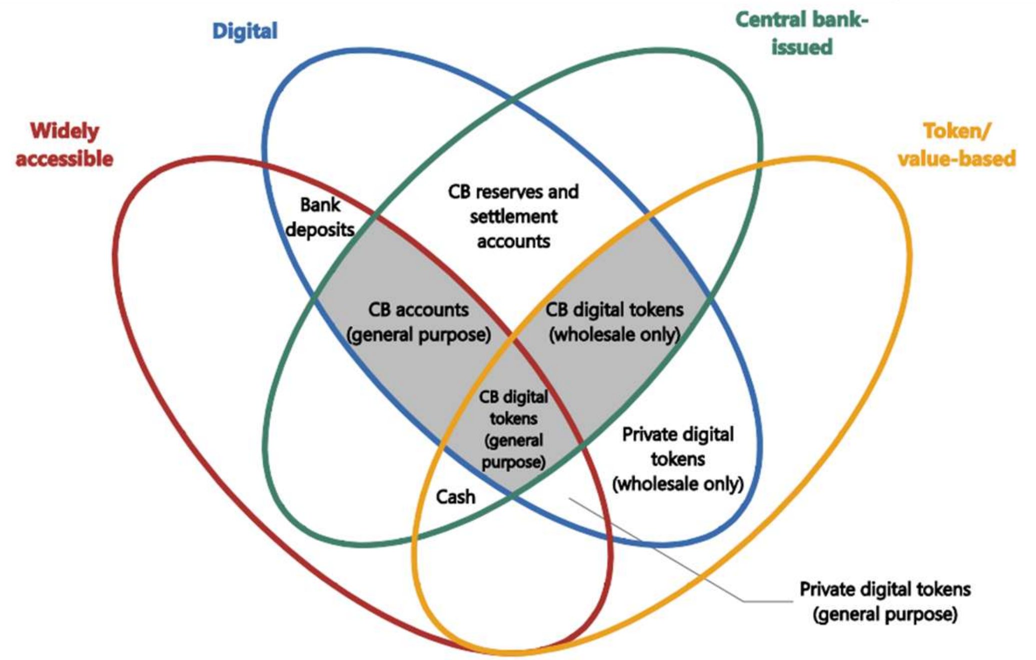
Payments Systems: Way to Transfer Value

- Either physically (cash, gold) or across accounts.
  - Cash or gold are tokens that represent value.
  - Transferring across accounts usually goes through a bank to settle
  - Payments are a lucrative business

" THE APPLE PAY DOESN'T WORK.... I'LL TRY A BANANA "

Types of Money

The money flower: a taxonomy of money   Graph 1



Source: BIS 2018 CPMI

Credit Cards

- The first credit card was introduced by the Bank of America in Fresno in 1950s
- In 1970, the BankAmericard system changed to a membership corporation and became VISA
- MasterCharge (later MasterCard) was started in California as a competitor to VISA
- Europay merged with MasterCard Intl. to form MasterCard Worldwide.

What drives CC profits?

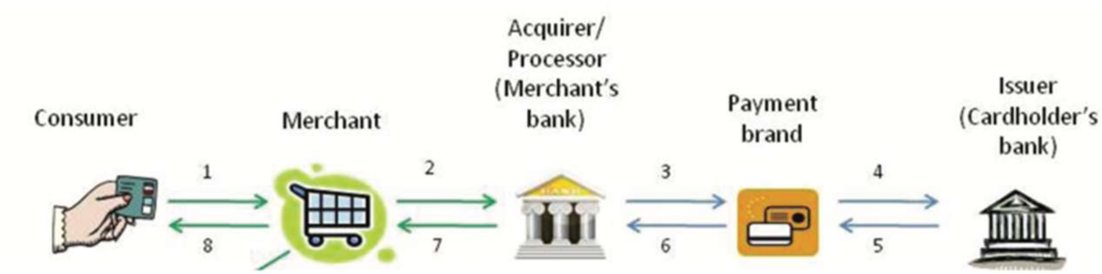
- APRs range from approx. 15%-25%
- At least 25% of card holders carry a balance.

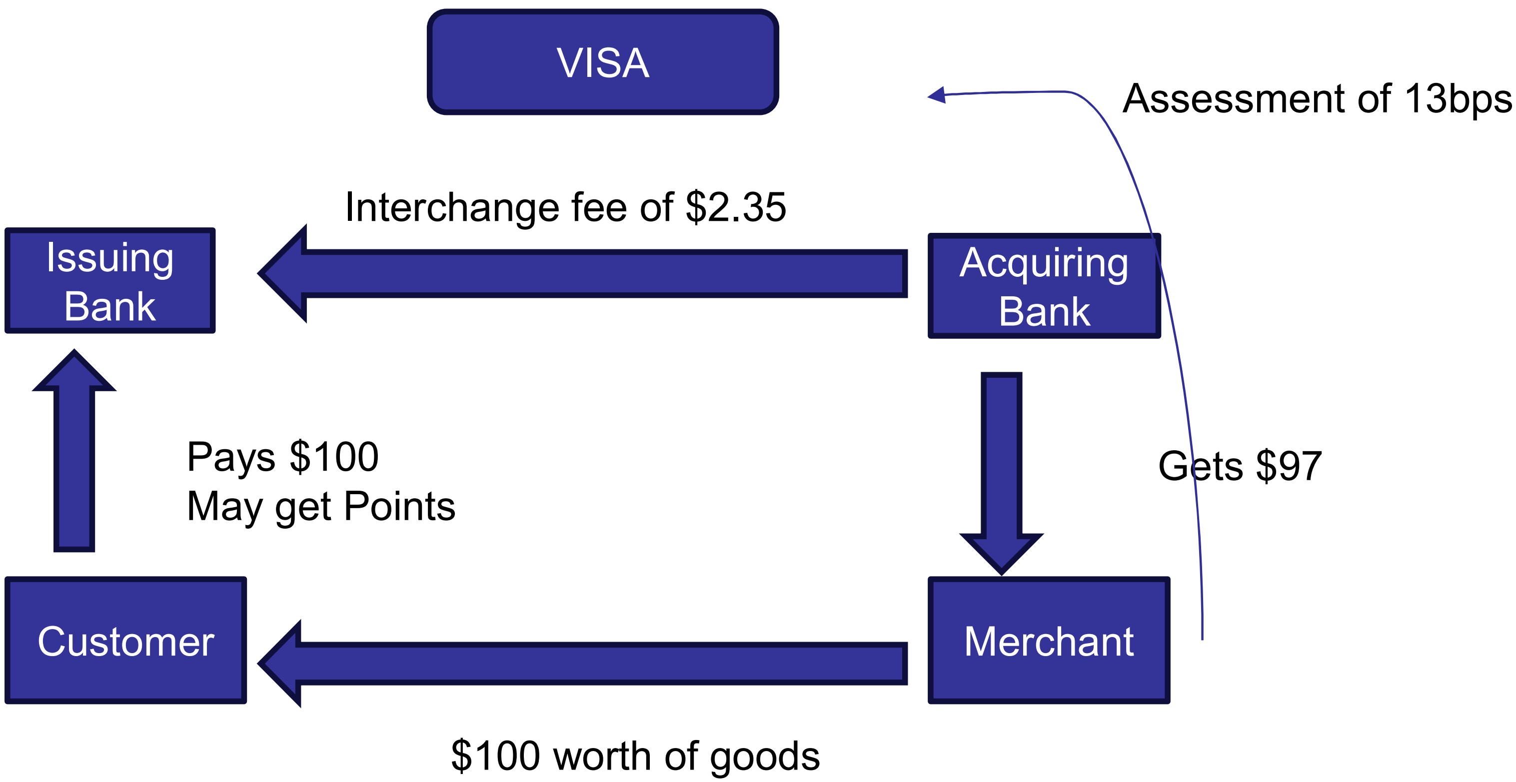
- Most states have usury laws (in CA 10%)
- In 1978 the Supreme Court ruled card issuers only subject to charter state laws
- CC operations moved to states with no usury laws.

- Credit cards are an example of a multi-sided market or platform
  1. Two retail sides: merchants and consumers
  2. Two banking sides: merchants’ banks and consumers’ banks.

- A card network – e.g., VISA is a platform that connects the two sides.
- The network makes revenue by charging fees to the users

What do Credit Card Networks do?




VISA
Assessment of 13bps
Interchange fee of $2.35
Issuing Bank
Acquiring Bank
Pays $100
May get Points
Gets $97
Customer
Merchant
$100 worth of goods

Definitions

Interchange Fee: The fee that the merchant's bank pays to the customer's bank.

- Provides an incentive for banks to issue credit cards
- Some countries have zero interchange fees e.g. Canada, Denmark, Finland
- Constant source of litigation

Merchant discount: The % of the transaction that the merchant pays to its bank.

- Discount depends on the type of purchase (e.g. alcohol versus food) and the size of the merchant (e.g. Walmart versus artisanal baker)

Assessment Fee: Fee paid by the merchants to the network

More on Interchange

- The interchange fee is set by payment networks
- Typically has two parts:
  1. A % of the transaction to the issuing bank e.g., 2.35%
  2. A fixed fee to the card network $0.15

- Varies based on card type (e.g., credit versus debit)
- How payment is made (in person, over the internet etc.)
- Industry and size of the merchant
- Region where the transaction occurs

Payment Processing is Profitable

- Visa net income is around 12 billion
- The company is not exposed to customer credit risk and makes no revenue on fees.
- Cash is anonymous, but payment processors know how money is spent.
- Electronic payments generate big data to sell for advertising.

High Costs

- Credit Card Fees mean that small transactions are not feasible.
- Stablecoins:
  - Cheaper settlement
  - Programmable

New Business Models: Agentic Finance

1. Agentic Commerce:
   - Automating search and purchases.
   - Naturally enabled with stablecoins (lower fees)

2. Agentic Treasury:
   - Reconciliation
   - Conditional payments

Agentic Commerce

- Automating search and purchase allows businesses to increase conversion
  1. Open Ai + Stripe ACP
  2. Visa's Agentic Finance
  3. Coinbasex402